Customer Churn Prediction Report

Executive Summary

This model can identify 90% of churners while misclassifying fewer than 1% of loyal customers, enabling highly targeted retention efforts with minimal outreach waste.

This project developed and evaluated machine learning models to predict customer churn and identify the key drivers influencing attrition risk. Among the models tested, the Random Forest classifier achieved best-in-class performance with:

- **Accuracy:** 96.8%
- **Precision:** 97.6%
- **Recall:** 90.2%
- **F1 Score:** 93.7%
- **ROC-AUC:** 0.992

The extremely high ROC-AUC indicates near-perfect class separation, meaning the model effectively distinguishes between churners and non-churners across decision thresholds.

Analysis of feature importance revealed that Customer Satisfaction Score is the dominant predictor of churn, significantly outweighing pricing, contract type, tenure, and usage variables. This suggests churn is driven primarily by customer experience factors rather than purely financial considerations.

Classification Threshold Analysis

While the model was evaluated using a standard probability threshold of 0.50, churn prediction is inherently a cost-sensitive problem. Adjusting the classification threshold allows the business to tradeoff between:

- Recall (capturing more churners)
- Precision (avoiding unnecessary retention outreach)

At the default 0.50 threshold:

- Recall: 90.2%
- False Positive Rate: <1%
- Only 5 non-churners were incorrectly flagged

If the threshold were lowered (e.g., to 0.40), recall would increase, capturing more at-risk customers, but false positives would rise. Conversely, increasing the threshold would reduce outreach costs but risk missing additional churners.

Because churn prevention typically has higher financial value than minor outreach costs, a slightly lower threshold may be strategically justified depending on retention campaign capacity and budget constraints.

Estimated Business Impact & ROI

To quantify potential value, we estimate financial impact using the model's test-set performance:

- Correctly identified churners (True Positives): 202
- Assume average Customer Lifetime Value (CLTV): $1,200
- Assume retention campaign success rate: 20%

Estimated retained customers:
202 × 20% = 40 customers

Estimated revenue preserved:
40 × $1,200 = $48,000

This estimate is based on a single test sample. Scaled across the full customer base, the potential annual retention impact could reach several hundred thousand dollars, depending on churn volume and campaign effectiveness.

Because false positives are extremely low (only 5 in the test set), outreach waste is minimal, making the retention strategy highly efficient.

Business Problem

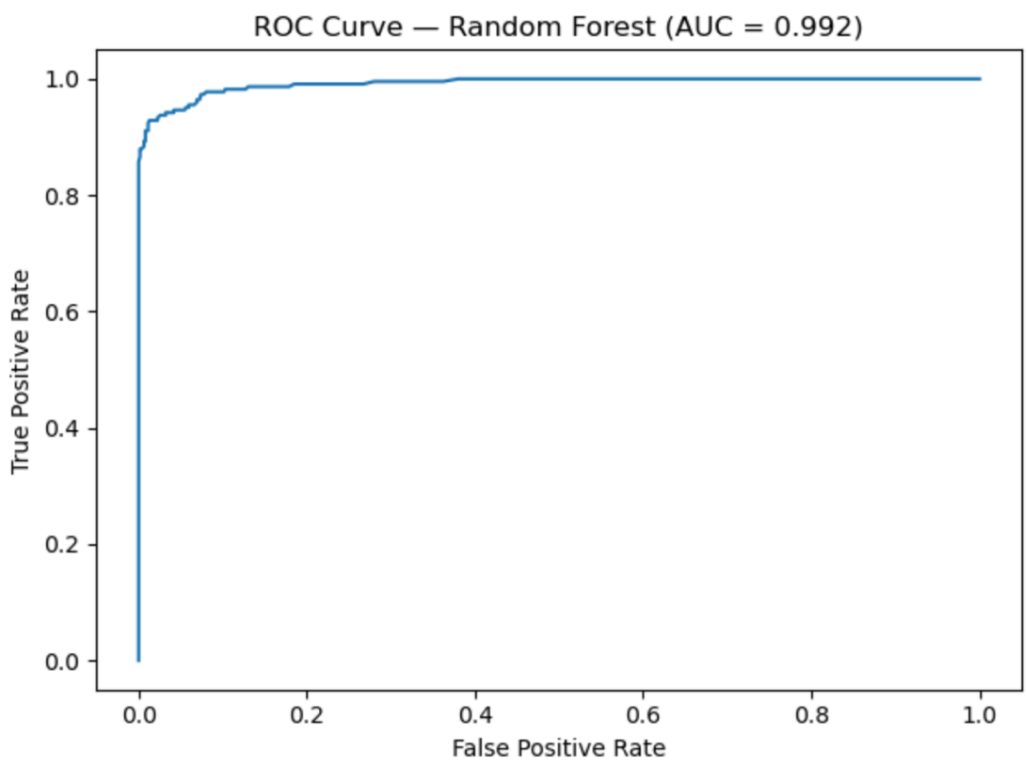
Customer churn reduces recurring revenue and increases acquisition costs. The objective was to build a predictive model capable of identifying customers at risk and quantifying the most influential churn drivers.

Model Comparison

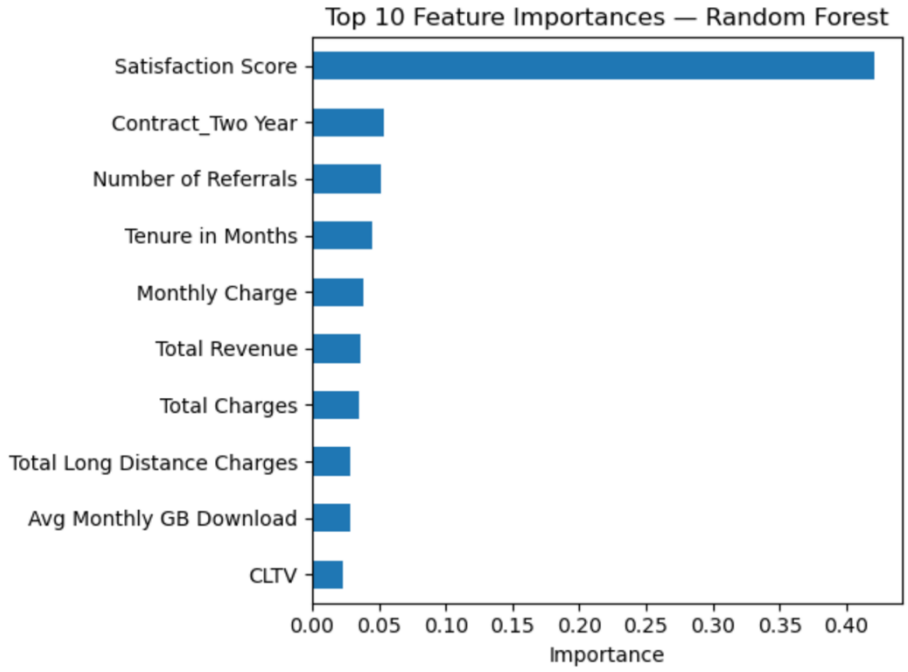
Bernoulli Naïve Bayes achieved 76.3% accuracy and demonstrated limited predictive capability. Logistic Regression (scaled) achieved 96.0% accuracy, 94% precision, 90.6% recall, and 92.3% F1 score. Random Forest outperformed both models, achieving the strongest accuracy, precision, F1 score, and ROC-AUC.

Model	Accuracy	Precision	Recall	F1
Bernoulli NB	0.763	0.543	0.683	0.605
Logistic Regression	0.960	0.940	0.906	0.923
Random Forest	0.968	0.976	0.902	0.937

Random Forest ROC Curve

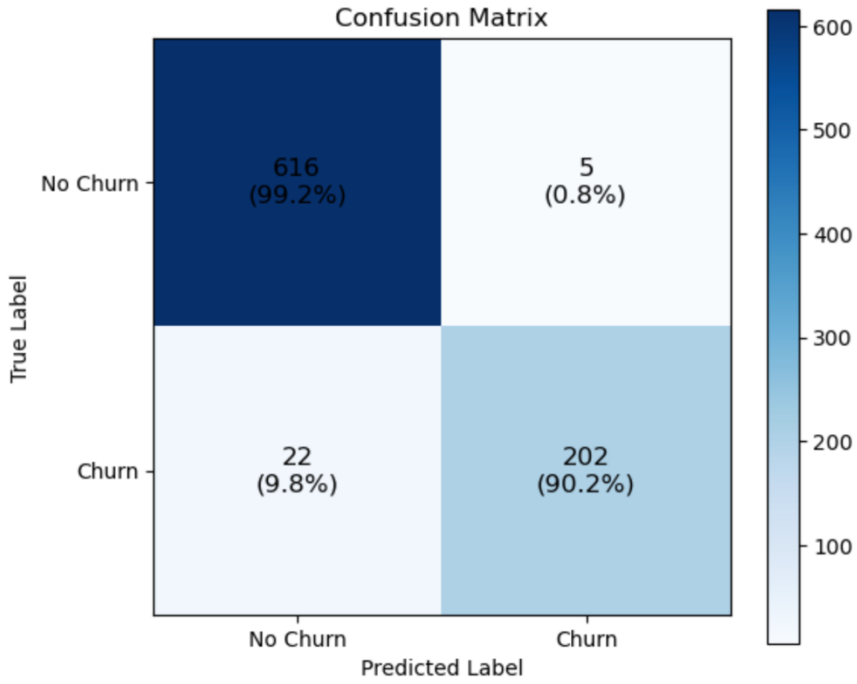


Random Forest Feature Importance Chart



Random Forest Confusion Matrix

True Negatives: 616 | False Positives: 5 | False Negatives: 22 | True Positives: 202



Key Drivers of Churn

Feature importance analysis revealed that Customer Satisfaction Score is the dominant predictor of churn, with significantly higher importance than contract type, tenure, or revenue metrics. This indicates that customer experience management is the most effective lever for churn reduction.

Business Recommendations

1. Prioritize Customer Experience Interventions

Feature importance analysis shows Customer Satisfaction Score is the strongest predictor of churn, significantly outweighing contract length, pricing, tenure, and usage metrics.

This indicates churn is primarily driven by customer experience rather than purely financial incentives.

Recommendation:

- Implement proactive outreach workflows for customers with low or declining satisfaction scores.
- Escalate unresolved support tickets tied to low satisfaction ratings.
- Monitor satisfaction trend deterioration as an early warning signal.

Improving customer experience should be the first-line defense against churn.

2. Deploy Risk-Based Retention Targeting

Using weekly or monthly churn scoring:

- Rank customers by predicted churn probability.
- Focus retention resources on the highest-risk decile.
- Align threshold selection with retention team capacity.

Given the model's high precision (97.6%), outreach waste is minimal, allowing highly efficient campaign targeting.

3. Optimize Threshold for Cost Sensitivity

Retention campaigns involve tradeoffs between:

- Missed churners (lost revenue)

- False positives (outreach cost)

The classification threshold should be tuned based on:

- Cost of churn
- Cost of retention intervention
- Campaign capacity constraints

A slightly lower threshold may be justified if the value of retained customers exceeds incremental outreach cost.

4. Integrate Churn Scoring into Operational Systems

For maximum impact:

- Embed model scoring into CRM systems.
- Trigger automated retention workflows.
- Track intervention success rates to refine ROI assumptions.
- Monitor performance drift quarterly.

Model deployment should be treated as a living system, not a one-time analysis.

Limitations

While the model demonstrates strong predictive performance, several limitations should be acknowledged:

1. Satisfaction Score Dominance

Satisfaction Score is the dominant predictor. While powerful, this variable may partially capture downstream effects (e.g., service failures, billing issues) rather than root causes.

Further segmentation analysis may be required to isolate underlying drivers of dissatisfaction.

2. Threshold Dominance

Model performance metrics were evaluated primarily at a 0.50 classification threshold. Different operational thresholds may alter precision-recall tradeoffs and should be calibrated against real retention costs.

3. Static Snapshot Evaluation

The model was evaluated on a static train-test split. Real-world deployment would require:

- Ongoing monitoring for data drift
- Periodic retraining
- Performance validation over time

4. ROI Estimates Are Assumption-Based

The ROI projection assumes:

- Average CLTV of $1,200
- 20% retention success rate

Actual financial impact will depend on campaign execution quality and customer response behavior.

A/B testing is recommended to validate real-world return.

Conclusion

This project successfully built a high-performing churn prediction model and identified customer satisfaction as the primary driver of churn. Improving customer experience is expected to generate the highest return on retention investment. If deployed and operationalized, this model provides a scalable data-driven retention strategy capable of materially improving recurring revenue.